



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

**FIRST SEMESTER 2026-
27COURSE HANDOUT**

Date: 02.08.2026

Course No : **EEE F437**
Course Title : **Semiconductor Fabrication Technology**
Pre-requisite of the Course : **Electronic Devices (F-214)**
Instructor-in-Charge : **Pankaj Arora (pankaj.arora@pilani.bits-pilani.ac.in)**
Practical Instructors : **Anita Khichar and Keerthana P.**

1. Course Description: The course describes both theoretical and practical aspects of semiconductor fabrication technology. This course will take you on the journey of silica sand to silicon integrated circuit (IC), i.e., “making a chip”. Converting a single crystal of silicon into an IC requires several fabrication steps, such as epitaxy, oxidation, chemical vapor deposition, metallization, ion implantation, diffusion, etching, and lithography. All these process steps will be discussed. Further, the modeling of each processing step would be discussed in detail.

2. Scope and Objective of the Course:

This course is designed to provide an introduction to the basic principles of semiconductor fabrication and to discuss various processing technologies. Fabrication of semiconductor chips is a joint venture by electrical engineers, chemical engineers, materials scientists, and physicists. This interdisciplinary course builds bridges across various disciplines.

3. Textbooks:

TB: J. Plummer, M. D. Deal, and P. B. Griffin, Silicon VLSI Technology: Fundamentals, Practice and Modeling, Pearson Education.

4. Reference Books:

R1: S. M. Sze, VLSI Technology, TATA McGraw-Hill.

R2: S.A. Campbell, The Science and Engineering of Microelectronic Fabrication, Oxford.

5. Course Plan:

Section	Lecture #	Topic	Brief	Reference
I	1 - 2	Introduction to the Course	A brief overview of the course and history, status, and future direction of IC fabrication.	Ch. 1 TB
II	3-5	Semiconductor Fabrication: Complete Process Flow	Fundamentals, Requirements, Perspectives, and Prospects	Ch. 2 TB
III	6- 8	Crystal structures, defects, directions, planes	Structural properties of Silicon	Ch. 3, TB
IV	9-11	Bulk single crystal growth to Wafer preparation, dopant distribution	To learn the art of wafer preparation and some of the basic properties of these wafers.	Ch. 3, TB



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

V	12	Semiconductor Manufacturing	Introduction to Clean room and wafer cleaning process (RCA)	Ch. 4, TB
VI	13-16	Oxidation and Si/SiO ₂ interface	Oxidation techniques and systems primarily focus on thermal oxidation and the properties at the Si/SiO ₂ interface.	Ch. 6, TB
VII	17-21	Lithography	Different lithography techniques (Photolithography, Electron beam, X-ray, etc.), learn how to print the patterns on the wafer using optical exposure systems. Basics of Optical Systems	Ch. 5, TB
VIII	22-25	Dopant Diffusion (Doping process)	Understand the doping concept by the diffusion method.	Ch. 7, TB
IX	26-29	Ion Implantation (Doping process)	Understand the doping concept by the Ion Implantation method	Ch. 8, TB
X	30	Annealing of damages and masking during implantation	How annealing helps recover damage, and how the thickness of the masking layer improves masking efficiency.	Ch. 8, TB
XI	31-34	Thin Film Deposition	Understand different techniques to deposit thin films (Chemical/Physical Vapor Deposition systems)	Ch. 9, TB
XII	35-38	Etching	Introducing various etching mechanisms involved in CMOS device fabrication	Ch. 10, TB
XIII	39-40	Metallization and Emerging Techniques	How devices are connected to the outside world, and Case Studies	Ch. 11, TB

6. Learning outcome of the course:

Students will understand the fundamental principles of semiconductor materials, crystal growth techniques, and wafer preparation used in IC fabrication. They will be able to describe and compare major semiconductor fabrication processes, including oxidation, diffusion, ion implantation, lithography, etching, and thin-film deposition. Students will be able to apply cleanroom practices, safety protocols, and contamination control methods relevant to semiconductor manufacturing. Finally, they will be able to correlate fabrication steps with the electrical characteristics of semiconductor devices.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

LIST OF EXPERIMENTS (tentative)

SI. No.	Experiment Name	Location
1.	To study/observe clean room formation, various components, and reliability.	CEERI Pilani
2.	Introduction to different substrates and the basics of the substrate cleaning process.	SMSD Lab (2116)
3.	Wafer dicing with a diamond-tip pen-scriber and a wafer dicing tool.	DST PURSE Lab
3.	Resistivity measurement of a semiconductor wafer using the 4-probe method.	Flexible Elect. Lab
4.	To study/observe XRD operation and how the crystallinity is measured. To study/observe XPS operation and how the chemical composition is measured.	CIF Facility
5.	Study/observe the UV-visible spectroscopy and calculate the optical bandgap Study/observe the Surface Morphology using Scanning Electron Microscopy Study/observe the characterization of graphene layers using Raman spectroscopy	CIF Facility
6.	Study/observe/understand the lithography operation	DST PURSE Lab
7.	Study/observe/understand the operation of different deposition tools.	DST PURSE Lab
8.	Light/Gas sensing measurement of semiconductor sensors	SMSD Lab (2116)
9.	Demonstration of the IV/CV Characterization using the Semiconductor Parameter Analyzer	CIF Facility
10.	One deposition experiment (complete process, from wafer loading to wafer pull out)	SMSD Lab (2116)

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 min.	30%	To be announced	CB
Quizzes	15 min.	15%	During lecture	CB
Regular Lab session and viva	----	15%	Regular	OB
Comprehensive exam.	120 min	40%	To be announced	OB + CB

8. Chamber Consultation Hour: To be announced in Class.

9. Notices: Notices for the course will be displayed on the EEE Notice Board and NALANDA.

10. Make-up Policy: Make-up will be allowed for genuine cases. A prior application should be sent to seek the same.

Instructor-in-charge
Course No. EEE F437